

CLASS 9th

MATHS REVISION GUIDE

Number System

- Rational, Irrational, Real numbers
- Laws of exponents: $a^m \cdot a^n = a^{m+n}$, $(a^m)^n = a^{mn}$

Algebra

- Factorization: $a^2 - b^2 = (a - b)(a + b)$, $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- Identities: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$

Geometry

- Pythagoras theorem: $a^2 + b^2 = c^2$
- Triangle types & properties, circles (chord, tangent, secant)

Coordinate Geometry

- Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint: $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$

Mensuration

- Cube: $SA = 6a^2, V = a^3$
- Sphere: $SA = 4\pi r^2, V = \frac{4}{3}\pi r^3$
- Cylinder: $SA = 2\pi r(h + r), V = \pi r^2 h$
- Cone: $SA = \pi r(l + r), V = \frac{1}{3}\pi r^2 h$

Statistics & Probability

- Mean: $\bar{x} = \frac{\sum x_i}{n}$
- Median: Middle value, Mode: Most frequent
- Probability: $P(E) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$

Linear Equations

- Standard: $ax + by + c = 0$, Solve by substitution/elimination/graphical method

Polygons

- Interior angles: $(n - 2) \cdot 180^\circ$
- Exterior angles: 360°